

Predictive Modelling of Mitral Valve Improvement Using Longitudinal Mixed-Effects Logistic Regression

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Introduction

Problem Description

Mitral valve disease can significantly impair cardiac function and often requires surgical intervention to restore normal valve performance. Two common surgical approaches are mini surgery and the conventional method; however, patient outcomes following surgery are heterogeneous. While some patients experience clear improvement in mitral valve functioning, others show limited or no improvement. Understanding whether mitral valve functioning improves over time, and whether improvement differs between surgical approaches, is therefore an important clinical and analytical problem.

From a data analytics perspective, this problem is challenging because patient outcomes are measured repeatedly over time. Each patient is observed at baseline and again at a 12-month follow-up, meaning observations within the same patient are correlated. Ignoring this longitudinal structure may lead to biased inference and misleading evaluation of predictive performance. In addition to inference, it is of interest to assess how well mitral valve improvement can be predicted for unseen patients and whether model performance is limited by underfitting or overfitting.

Research questions

1. Does mitral valve functioning improve from baseline to 12 months following surgical intervention?
2. Does the longitudinal pattern of improvement differ between mini surgery and conventional surgery?
3. How accurately can mitral valve improvement (Yes/No) be predicted using available patient and treatment information?
4. How does predictive performance differ between standard logistic regression and mixed-effects logistic regression, and what does this reveal about underfitting, overfitting, and the bias-variance trade-off?

Data and Problem Summary

The analysis is based on a real-world healthcare dataset containing information on 1000 patients who underwent mitral valve surgery. Each patient was observed at two time points: baseline and a 12-month post-surgery follow-up. The primary outcome variable records whether mitral valve functioning improved, measured as a binary response (Yes/No).

The dataset includes:

- `Subject` (patient ID)
- `Surgery` (Mini / Conventional)
- `Gender` (Male / Female)
- `Smoke` (Yes / No)
- `Mitral.valve.improved.baseline` (Yes / No)
- `Mitral.valve.improved.post.surgery` (Yes / No)

Because each patient is measured at both baseline and 12 months, the data have a longitudinal structure with repeated observations nested within individuals. The outcome variable—mitral valve functioning improved (Yes/No)—is binary and is therefore modelled using logistic regression. A standard logistic regression model is used as a baseline but assumes independence between observations, while a mixed-effects logistic regression incorporates patient-level random effects to account for within-patient correlation. Comparing these approaches allows assessment of whether modelling patient heterogeneity improves inference and predictive performance.

Brief Related Work

Predictive modelling techniques are widely used in healthcare research to analyse treatment effectiveness and patient outcomes. Prior studies highlight that longitudinal data provide richer information than single time-point analyses, as they allow changes within individuals to be distinguished from between-patient variability (Fitzmaurice, Laird and Ware, 2011). Research also shows that models which ignore within-subject dependence can produce biased estimates and unreliable performance assessment.

Mixed-effects and multilevel models are commonly recommended for analysing longitudinal medical data, particularly when outcomes are binary and repeatedly measured. These models allow population-level effects and patient-specific heterogeneity to be modelled simultaneously, improving inferential validity and stabilising estimates through partial pooling (Verbeke and Molenberghs, 2000). Motivated by this literature, the present study applies mixed-effects logistic regression and evaluates predictive performance to assess bias, variance, and generalisation ability in the context of mitral valve surgery outcomes.

Methodology

Overview of Predictive Modelling Approach

This study adopts a predictive analytics framework to model and evaluate mitral valve improvement following surgery using longitudinal patient data. Because the outcome variable—mitral valve functioning improved—is binary, logistic regression-based methods are used throughout. Two modelling approaches are explored and compared:

- Standard logistic regression (GLM) as a baseline predictive model, assuming independence between observations.
- Mixed-effects logistic regression (GLMM), incorporating patient-level random effects to explicitly model within-patient correlation arising from repeated measurements.

The analysis proceeds in a structured pipeline consisting of data preparation, model specification, model selection, and model validation, with particular attention to bias–variance trade-offs, overfitting, and underfitting.

I also tested whether adding an interaction term (Surgery $\times$ Time) improves model fit.

Model Specification

Baseline Model: Standard Logistic Regression (GLM)

As a reference point, a standard logistic regression model was fitted using surgery type and time as predictors, with and without an interaction term. These models provide population-averaged estimates and serve as a baseline for comparison, but they assume independence between observations—an assumption violated in longitudinal data.

The interaction model allows assessment of whether the effect of time differs by surgery type, addressing potential differential recovery trajectories.

Mixed-Effects Logistic Regression (GLMM)

To properly account for repeated measurements within patients, mixed-effects logistic regression models were fitted with a random intercept for patient ID. This random effect captures unobserved patient-specific heterogeneity and reduces variance through partial pooling. Both a main-effects model and an interaction model (Surgery $\times$ Time) were evaluated.

Model selection between these specifications was conducted using likelihood ratio tests and information criteria (AIC), balancing goodness of fit against model complexity.

Model Selection

Model selection focused on determining whether increased model complexity meaningfully improved explanatory power. Likelihood ratio testing showed that including the Surgery $\times$ Time interaction did not significantly improve model fit relative to the main-effects mixed model. Estimated odds ratios for fixed effects were close to one and statistically non-significant, indicating weak effect sizes.

Based on these results, the simpler mixed-effects model was preferred for inference, as it adequately accounted for within-patient correlation while avoiding unnecessary complexity that could increase variance.

Model Validation and Predictive Performance

To evaluate predictive performance fairly, a patient-level 70/30 train–test split was used, ensuring that all observations from a given patient were assigned to either the training or test set, thereby avoiding information leakage.

Both the GLM and GLMM were trained on the same training data and evaluated on unseen patients. Predictive performance was assessed using multiple complementary metrics:

- ROC curves and AUC to evaluate discrimination
- Brier score to assess probabilistic accuracy
- Confusion matrix-based metrics (accuracy, sensitivity, specificity, F1-score) at a 0.5 threshold

ROC analysis showed AUC values close to 0.5 for both models, indicating weak discrimination. Confusion matrices revealed moderate specificity but low sensitivity, meaning the models were better at identifying non-improvement than improvement.

Threshold Tuning and Diagnostic Analysis

To explore whether performance was sensitive to the threshold, threshold tuning was conducted over a grid of values from 0.05 to 0.95. Accuracy and F1-score were computed at each threshold for both models. Although lower thresholds increased F1-scores, this was achieved by predicting nearly all cases as “improved,” leading to collapsed specificity. This confirmed that apparent gains were artefacts of threshold choice rather than genuine predictive improvement.

Bias / variance / overfitting / underfitting

- Bias: happens if the model is too simple (e.g., missing Surgery × Time when it matters).
- Variance: happens when predictions change a lot across different training samples.
- Overfitting: model fits training data well but performs worse on unseen patients.
- Underfitting: model is too simple and cannot capture patterns in the data.

Several diagnostic procedures were used to assess these behaviours. Training–test error curves across increasing training sizes showed persistently high and closely aligned errors, indicating high bias and underfitting rather than overfitting. Repeated random patient-level splits revealed variability in test error, reflecting sampling variance due to patient heterogeneity. However, the absence of a large gap between training and test performance suggests that overfitting is limited. In theory, the standard logistic regression (GLM) is biased due to its independence assumption, while the mixed-effects model (GLMM) reduces variance by accounting for within-patient correlation. The near-identical predictive performance of the two models indicates that the dominant limitation is structural bias caused by weak predictive signal in the available features, rather than model misspecification or excessive complexity.

importing libraries

```
#installing and importing libraries needed  
library(stats)  
library(ggplot2)  
library(caret)
```

```
## Loading required package: lattice
```

```
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
library(nlme)
```

```
##  
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:lme4':  
##  
##      lmList
```

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##  
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':  
##  
##      cov, smooth, var
```

loading the data set and previewing the first few rows

```
#loading dataset - looking at the basic data in the csv file  
data <- read.csv("Mitral valve(in).csv")  
  
head(data) # previewing first few rows
```

```
##      X Subject      Surgery Gender Smoke Mitral.valve.improved.baseline
## 1 1      1      Mini Female   Yes      Yes
## 2 2      2 Conventional Female   No      No
## 3 3      3 Conventional Female   Yes     No
## 4 4      4 Conventional Female   Yes     No
## 5 5      5 Conventional   Male   Yes     Yes
## 6 6      6 Conventional   Male   No      No
## Mitral.valve.improved.post.surgery
## 1      Yes
## 2      Yes
## 3      No
## 4      Yes
## 5      No
## 6      No
```

```
summary(data) # displays the summary of the data
```

```
##      X      Subject      Surgery      Gender
## Min.   : 1.0   Min.   : 1.0   Length:1000   Length:1000
## 1st Qu.: 250.8 1st Qu.: 250.8   Class :character   Class :character
## Median : 500.5 Median : 500.5   Mode  :character   Mode  :character
## Mean    : 500.5 Mean    : 500.5
## 3rd Qu.: 750.2 3rd Qu.: 750.2
## Max.    :1000.0 Max.    :1000.0
## Smoke      Mitral.valve.improved.baseline
## Length:1000 Length:1000
## Class :character   Class :character
## Mode  :character   Mode  :character
##
##
## Mitral.valve.improved.post.surgery
## Length:1000
## Class :character
## Mode  :character
##
##
##
```

```
str(data) #displays the structer of the data set
```

```
## 'data.frame':    1000 obs. of  7 variables:
## $ X                : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Subject          : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Surgery          : chr  "Mini" "Conventional" "Conventional"
## $ Gender           : chr  "Female" "Female" "Female" "Female"
## $ Smoke            : chr  "Yes" "No" "Yes" "Yes" ...
## $ Mitral.valve.improved.baseline : chr  "Yes" "No" "No" "No" ...
## $ Mitral.valve.improved.post.surgery: chr  "Yes" "Yes" "No" "Yes" ...
```

```
names(data) # displays the column names of the data set
```

```
## [1] "X"                "Subject"
## [3] "Surgery"          "Gender"
## [5] "Smoke"            "Mitral.valve.improved.baseline"
## [7] "Mitral.valve.improved.post.surgery"
```

```
colSums(is.na(data)) # checks for missing values in the columns from the dataset
```

```
##                X                Subject
##                0                0
##                Surgery            Gender
##                0                0
##                Smoke    Mitral.valve.improved.baseline
##                0                0
## Mitral.valve.improved.post.surgery
##                0
```

Data reshaping and preparation and Feature Engineering

The dataset is initially provided in wide format, with separate outcome variables for baseline and 12-month follow-up. To correctly model the longitudinal structure of the data, it is reshaped into long format, where each row represents a single patient–time observation. This person–period representation is required for mixed-effects modelling and ensures that within-patient correlation across repeated measurements is explicitly captured.

Defining a patient identifier (id) enables the inclusion of patient-level random effects, which reduces bias arising from unmodelled heterogeneity and helps control variance through partial pooling. Ordering the data by patient facilitates verification of the longitudinal structure and supports reliable model fitting and validation.

```
# Reshaping the data from wide to long format
data=reshape(data,v.names = "Improved",
              varying = c("Mitral.valve.improved.baseline","Mitral.valve.improved.p
ost.surgery"),
              timevar = "Time",
              times = c("Baseline","Month12"),
              direction = "long")

data$id=data$Subject # patient id/ identifier

data=data[order(data$Subject),] # ordering data by Subject to keep it tidy/identif
er
head(data) # checking the data after reshaping
```

```
##           X Subject      Surgery Gender Smoke      Time Improved id
## 1.Baseline 1         1         Mini Female   Yes Baseline      Yes  1
## 1.Month12  1         1         Mini Female   Yes Month12      Yes  1
## 2.Baseline 2         2 Conventional Female   No  Baseline      No   2
## 2.Month12  2         2 Conventional Female   No  Month12      Yes  2
## 3.Baseline 3         3 Conventional Female   Yes Baseline      No   3
## 3.Month12  3         3 Conventional Female   Yes Month12      No   3
```

Creating binary outcome variable for binomial mixed models (0/1)

This code block cleans and the response variable was converted into a binary indicator (Improved01), coded as 1 for improvement and 0 otherwise, to allow binomial logistic modelling. It converts all categorical predictors to factors so coefficients are estimated relative to defined reference levels. Setting Time = "Baseline" as the reference ensures time effects are interpreted as changes from baseline, and the frequency table confirms the longitudinal balance across time points. No additional feature engineering was performed, as the focus of the study was on evaluating modelling structure rather than expanding the predictor space.

```
data$Improved=trimws(as.character(data$Improved)) # clean response text

data$Improved01=ifelse(toupper(data$Improved)=="YES",1,0) # binary variable for logistic regression - Creates a 0/1 binary outcome where "YES" is coded as 1 and everything else is coded as 0

#Binary coding enables logistic regression modelling; predictors are set as factors with Baseline as reference for time.

# predictors as factors
data$Surgery = as.factor(data$Surgery)
data$Time = as.factor(data$Time)
data$Gender = as.factor(data$Gender)
data$Smoke = as.factor(data$Smoke)

data$Time = relevel(data$Time, ref = "Baseline") # set Baseline as reference

table(data$Time) #Checks the number of observations at each time point to confirm both levels are present after preprocessing.
```

```
##
## Baseline  Month12
##      1000      1000
```

Measurements within the same patient appear correlated, suggesting that a multilevel modelling approach is appropriate.

After reshaping the data into long format, this code produces frequency tables for smoking status, the binary outcome variable, and gender to verify that these variables were retained correctly across repeated observations and to check their distribution before modelling.

```
table(data$Smoke)
```

```
##
##   No  Yes
##  984 1016
```

```
table(data$Improved01)
```

```
##
##    0    1
## 1035  965
```

```
table(data$Gender)
```

```
##
## Female    Male
##    1008    992
```

clean data set

This code displays the structure and first few rows of the processed/clean dataset to verify variable types and the longitudinal format after preprocessing. It then saves the cleaned dataset as a CSV file for reproducibility and use in subsequent modelling steps.

```
# Preview of the clean longitudinal dataset
str(data)
```

```
## 'data.frame':    2000 obs. of  9 variables:
## $ X          : int  1 1 2 2 3 3 4 4 5 5 ...
## $ Subject    : int  1 1 2 2 3 3 4 4 5 5 ...
## $ Surgery    : Factor w/ 2 levels "Conventional",...: 2 2 1 1 1 1 1 1 1 1 ...
## $ Gender     : Factor w/ 2 levels "Female","Male": 1 1 1 1 1 1 1 1 2 2 ...
## $ Smoke      : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 2 2 2 2 2 ...
## $ Time       : Factor w/ 2 levels "Baseline","Month12": 1 2 1 2 1 2 1 2 1 2 ...
## $ Improved   : chr  "Yes" "Yes" "No" "Yes" ...
## $ id         : int  1 1 2 2 3 3 4 4 5 5 ...
## $ Improved01: num  1 1 0 1 0 0 0 1 1 0 ...
## - attr(*, "reshapeLong")=List of 4
## ..$ varying:List of 1
## .. ..$ Improved: chr [1:2] "Mitral.valve.improved.baseline" "Mitral.valve.improved.post.surgery"
## .. ..- attr(*, "v.names")= chr "Improved"
## .. ..- attr(*, "times")= chr [1:2] "Baseline" "Month12"
## ..$ v.names: chr "Improved"
## ..$ idvar   : chr "id"
## ..$ timevar : chr "Time"
```

```
head(data)
```

```
##           X Subject      Surgery Gender Smoke      Time Improved id Improved01
## 1.Baseline 1         1         Mini Female   Yes Baseline      Yes    1         1
## 1.Month12  1         1         Mini Female   Yes Month12      Yes    1         1
## 2.Baseline 2         2 Conventional Female    No Baseline      No    2         0
## 2.Month12  2         2 Conventional Female    No Month12      Yes    2         1
## 3.Baseline 3         3 Conventional Female   Yes Baseline      No    3         0
## 3.Month12  3         3 Conventional Female   Yes Month12      No    3         0
```

```
# Saves clean dataset
write.csv(data, "clean dataset/Mitral_valve_clean.csv", row.names = FALSE)
```

Results

Descriptive summaries

Descriptive summaries were used to assess group balance across surgical approaches, the distribution of observations across time points, outcome prevalence, and the presence of missing data. Crude improvement rates by surgery and time provide preliminary insight into potential differences in outcomes prior to formal modelling prior to fitting longitudinal models.

```
table(data$Surgery)
```

```
##
## Conventional      Mini
##           1002      998
```

Creates a frequency table showing how many patients are in each surgical group (Mini vs Conventional). I used this as this checks group balance. Large imbalances can introduce bias and affect model stability.

```
table(data$Time)
```

```
##
## Baseline  Month12
##       1000      1000
```

#Counts the number of observations at each time point (Baseline and Month12).I used this as this confirms that each patient contributes repeated measurements and helps identify drop-out or missing follow-ups.

```
table(data$Improved01)
```

```
##
##      0      1
## 1035   965
```

Shows how many observations correspond to improvement (1) and no improvement (0).I used this as this assesses the distribution/class balance of the binary outcome, which is important for model fitting and interpretation. Severe imbalance can affect prediction accuracy and model calibration.

```
colSums(is.na(data))
```

```
##           X      Subject      Surgery      Gender      Smoke      Time      Improved
##           0          0          0          0          0          0          0
##           id Improved01
##           0          0
```

Calculates the total number of missing values in each column of the data set. I used this as this helps identify any missing data issues that may need to be addressed before analysis, such as through imputation or sensitivity analyses.

Descriptive improvement rates by surgery and time prior to modelling

This code computes and visualises mean improvement proportions by surgery type and time point as an exploratory step. By examining group-level trends before modelling, it helps assess whether simple patterns exist in the data or whether substantial unexplained variability remains, providing early insight into potential bias (underfitting) and variance (instability) that the subsequent models need to address.

```
aggregate(Improved01 ~ Surgery + Time, data=data, mean) # improvement rate by surgery and time
```

```
##           Surgery      Time Improved01
## 1 Conventional Baseline  0.4690619
## 2           Mini Baseline  0.5230461
## 3 Conventional Month12   0.4750499
## 4           Mini Month12  0.4629259
```

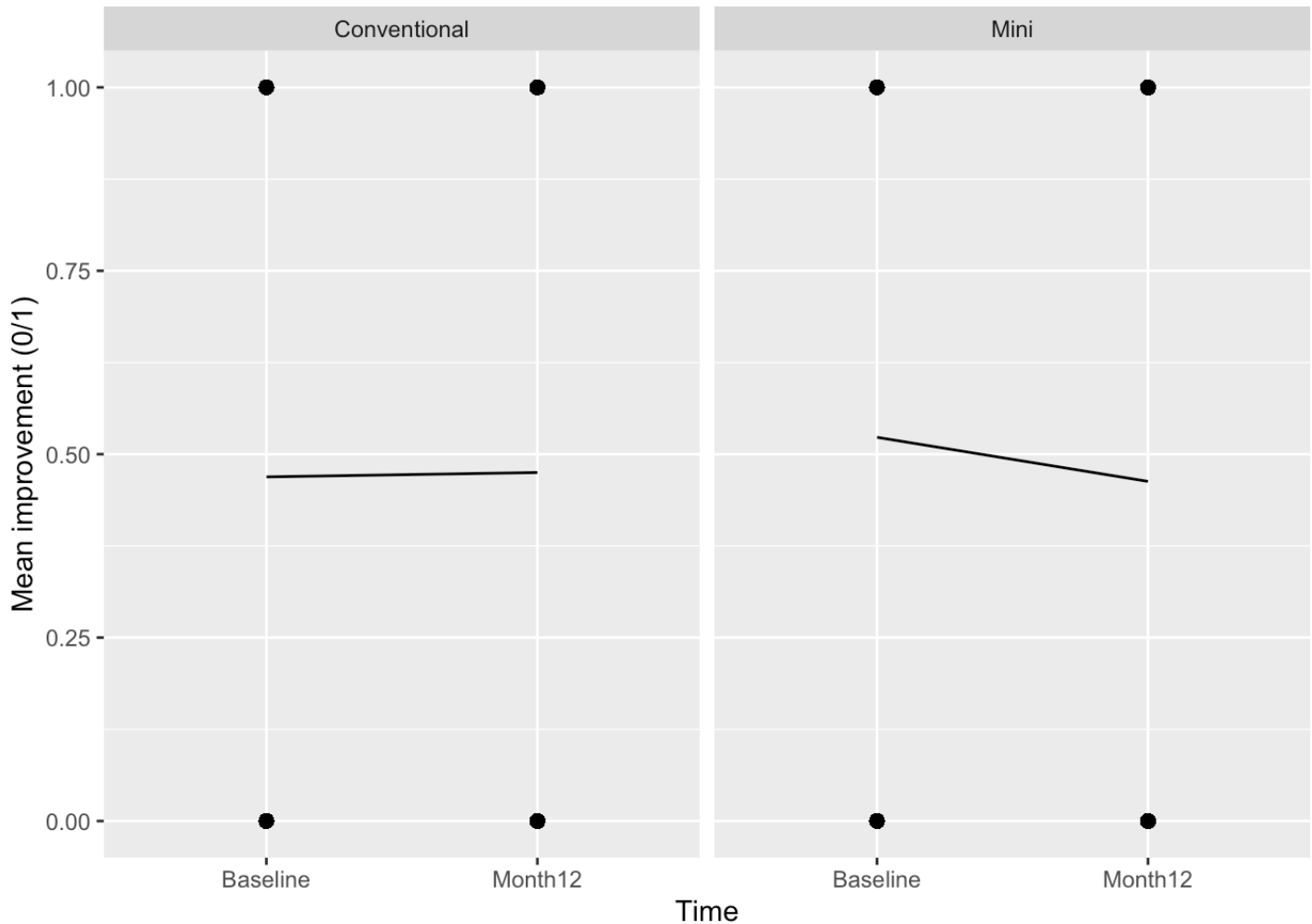
Computes the mean of Improved01 for each combination of Surgery and Time, because Improved01 is coded as 0/1, the mean represents the proportion of patients. Provides an interpretable descriptive summary of improvement rates by:

```
#surgical approach
#time point
```

#This helps contextualise model results and supports interpretation of interaction effects.

#Visualises average change in improvement probability over time within each surgical group; motivates testing Surgery × Time interaction.

```
# plot mean improvement by time, split by surgery
ggplot(data, aes(Time, Improved01, group = id)) +
  stat_summary(fun = mean, geom = "point", size = 2) +
  stat_summary(fun = mean, geom = "line", aes(group = 1)) +
  facet_wrap(~Surgery) +
  ylab("Mean improvement (0/1)") +
  xlab("Time")
```



Interpretation of descriptive improvement rates by surgery and time

The descriptive improvement rates show only small differences between surgery types and minimal change over time, with mean proportions remaining close to 0.5 across both baseline and 12 months. This weak separation suggests that surgery and time alone provide limited explanatory power, indicating **potential high bias (underfitting)** if modelled without additional structure. At the same time, the shallow group-level trends relative to individual variability **imply high variance**, motivating the use of a **mixed-effects logistic regression model to reduce variance** through patient-level random effects while formally testing the Surgery $\times$ Time interaction.

Standard logistic regression (GLM baseline)

Main effects model (Model)

This code fits standard logistic regression models with main effects only (Model 1) and with a Surgery $\times$ Time interaction as a baseline comparison, under the assumption of independent observations. These GLM models provide population-level estimates and odds ratios that help assess whether surgery and time alone capture meaningful signal; however, because they ignore within-patient correlation, they are

expected to exhibit structural bias and potentially unstable estimates (higher variance) in longitudinal data. The GLM results therefore serve as a reference point for evaluating whether the mixed-effects model reduces bias and variance by accounting for patient-level heterogeneity.

```
# GLM baseline (assumes independence) - used as a comparison only
m_glm1=glm(Improved01~Surgery+Time,family="binomial",data=data)
summary(m_glm1)
```

```
##
## Call:
## glm(formula = Improved01 ~ Surgery + Time, family = "binomial",
##      data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.05787    0.07746  -0.747   0.455
## SurgeryMini  0.08390    0.08955   0.937   0.349
## TimeMonth12 -0.10821    0.08955  -1.208   0.227
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2770.1  on 1999  degrees of freedom
## Residual deviance: 2767.8  on 1997  degrees of freedom
## AIC: 2773.8
##
## Number of Fisher Scoring iterations: 3
```

```
exp(coef(m_glm1))
```

```
## (Intercept) SurgeryMini TimeMonth12
##      0.9437711    1.0875162    0.8974423
```

```
m_glm2=glm(Improved01~Surgery*Time,family="binomial",data=data)
summary(m_glm2)
```

```
##
## Call:
## glm(formula = Improved01 ~ Surgery * Time, family = "binomial",
##      data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.12391    0.08952  -1.384   0.1663
## SurgeryMini      0.21616    0.12668   1.706   0.0879 .
## TimeMonth12      0.02403    0.12656   0.190   0.8494
## SurgeryMini:TimeMonth12 -0.26485    0.17920  -1.478   0.1394
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 2770.1  on 1999  degrees of freedom
## Residual deviance: 2765.6  on 1996  degrees of freedom
## AIC: 2773.6
##
## Number of Fisher Scoring iterations: 3
```

```
exp(coef(m_glm2))
```

```
##              (Intercept)              SurgeryMini              TimeMonth12
##              0.8834586                1.2413016                1.0243184
## SurgeryMini:TimeMonth12
##              0.7673238
```

Interpretation of data structure and GLM results

The data summaries confirm that the longitudinal dataset is well-structured after reshaping, with exactly two observations per patient (baseline and month 12), balanced group sizes across surgery types and time points, and no missing values. Despite this balance, both the descriptive improvement rates and the standard logistic regression results show only weak differences between surgery types and minimal change over time, with odds ratios close to one and no statistically significant main or interaction effects. This indicates high bias (underfitting), as surgery and time alone explain little variation in the outcome. At the same time, the large residual deviance relative to the null deviance and the absence of patient-level structure suggest that substantial unexplained heterogeneity remains, contributing to high variance when independence is assumed. These findings motivate the use of a mixed-effects logistic regression model to reduce variance by accounting for within-patient correlation and to assess whether subject-specific effects improve model fit and inference beyond the population-averaged GLM.

Interaction model (Model 2): mixed-effects logistic regression

This code fits mixed-effects logistic regression models with a patient-level random intercept to account for within-patient correlation in the longitudinal binary outcome. By including a random intercept, the model reduces variance through partial pooling and mitigates bias introduced by the independence assumption

in the GLM. The interaction model allows the effect of time to differ by surgery type, formally testing whether improvement trajectories vary across surgical approaches while avoiding overfitting by keeping the random-effects structure parsimonious.

```
#Estimates effects of surgery and time while accounting for patient-level heterogeneity using a random intercept.
```

```
#Fitting a multilevel longitudinal model (binary outcome => logistic mixed model).
```

```
# Mixed-effects logistic regression
```

```
m1=glmer(Improved01~Surgery+Time+(1|id),data=data,family=binomial)
summary(m1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
```

```
## Approximation) [glmerMod]
```

```
## Family: binomial ( logit )
```

```
## Formula: Improved01 ~ Surgery + Time + (1 | id)
```

```
## Data: data
```

```
##
```

```
##      AIC      BIC    logLik -2*log(L)  df.resid
##  2774.6   2797.0   -1383.3   2766.6     1996
```

```
##
```

```
## Scaled residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -1.0142 -0.9253 -0.8753  0.9907  1.0859
```

```
##
```

```
## Random effects:
```

```
## Groups Name      Variance Std.Dev.
```

```
## id      (Intercept) 0.1071  0.3273
```

```
## Number of obs: 2000, groups: id, 1000
```

```
##
```

```
## Fixed effects:
```

```
##      Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept) -0.05944    0.07987  -0.744    0.457
```

```
## SurgeryMini  0.08618    0.09312   0.925    0.355
```

```
## TimeMonth12 -0.11116    0.09081  -1.224    0.221
```

```
##
```

```
## Correlation of Fixed Effects:
```

```
##      (Intr) SrgryM
```

```
## SurgeryMini -0.582
```

```
## TimeMonth12 -0.566 -0.002
```

```
# Mixed-effects logistic regression with interaction
```

```
m2=glmer(Improved01~Surgery*Time+(1|id),data=data,family=binomial)
```

```
summary(m2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: Improved01 ~ Surgery * Time + (1 | id)
## Data: data
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##      2774.3      2802.3    -1382.2    2764.3      1995
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.0494 -0.9272 -0.8800  1.0086  1.0785
##
## Random effects:
## Groups Name          Variance Std.Dev.
## id      (Intercept) 0.1101    0.3318
## Number of obs: 2000, groups: id, 1000
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.12739    0.09203  -1.384    0.166
## SurgeryMini      0.22223    0.13028   1.706    0.088 .
## TimeMonth12      0.02470    0.12833   0.192    0.847
## SurgeryMini:TimeMonth12 -0.27228    0.18183  -1.497    0.134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) SrgryM TmMn12
## SurgeryMini -0.707
## TimeMonth12 -0.698  0.493
## SrgryM:TM12  0.494 -0.698 -0.706
```

How Surgery changes over the time (interaction) as this allows the time effect to differ by surgery type, addressing whether improvement trajectories vary across approaches.

Interpretation of mixed-effects interaction model (Model 2)

The mixed-effects models confirm that accounting for patient-level heterogeneity introduces a non-negligible random-intercept variance, indicating substantial between-patient variability that was ignored in the GLM and therefore contributing to high variance under the independence assumption. Despite this variance reduction through partial pooling, the fixed effects for surgery, time, and their interaction remain statistically non-significant and have odds ratios close to one, with only a marginal indication of a surgery effect in the interaction model. The minimal improvement in AIC relative to the main-effects mixed model suggests that allowing time effects to differ by surgery adds limited explanatory power.

Together, these results indicate that while the mixed-effects framework appropriately controls variance and avoids overfitting, structural bias (underfitting) remains because surgery and time alone do not sufficiently explain outcome variation. This aligns with the descriptive and GLM findings and suggests that additional covariates or longer follow-up would be required to meaningfully reduce bias and improve predictive performance.

Model comparison + Odds ratios

This code performs a likelihood ratio test to formally compare the main-effects mixed model with the interaction model, assessing whether adding the Surgery $\times$ Time term meaningfully improves model fit or merely increases complexity. The odds ratios from the interaction model are then reported to aid interpretation of effect sizes; together, these steps help determine whether the interaction reduces bias (underfitting) without introducing unnecessary complexity that could increase variance or overfitting.

```
anova(m1,m2,test="Chisq") # Likelihood ratio test comparing main effects vs interaction model
```

```
## Data: data
## Models:
## m1: Improved01 ~ Surgery + Time + (1 | id)
## m2: Improved01 ~ Surgery * Time + (1 | id)
##      npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m1      4 2774.6 2797.0 -1383.3   2766.6
## m2      5 2774.3 2802.3 -1382.2   2764.3 2.2462  1      0.1339
```

```
exp(fixef(m2)) # Odds ratios for fixed effects in interaction model
```

```
##              (Intercept)              SurgeryMini              TimeMonth12
##              0.8803934              1.2488610              1.0250063
## SurgeryMini:TimeMonth12
##              0.7616408
```

Interpretation of model comparison and odds ratios

The likelihood ratio test comparing the main-effects and interaction mixed-effects models shows that adding the Surgery $\times$ Time interaction does not significantly improve model fit ($\chi^2 = 2.25$, $p = 0.134$), despite a very small reduction in AIC. This indicates that the interaction term adds complexity without meaningfully reducing bias (underfitting) and therefore risks unnecessary model complexity. The estimated odds ratios from the interaction model are all close to one, suggesting weak effect sizes and reinforcing that surgery type and time explain limited variation in the outcome. Overall, **retaining the simpler main-effects mixed model is preferable**, as it adequately controls variance through random effects while avoiding overfitting driven by an unsupported interaction term.

Longitudinal clustering of binary outcomes by patient

This code visualises repeated binary outcomes for each patient across time by plotting individual trajectories of improvement, with points jittered to reduce overlap. The clear clustering of observations within patients highlights within-subject correlation, **justifying the use of a mixed-effects model to**

reduce variance and avoid biased inference that would arise from assuming independence.

```
#Shows within-patient similarity across time, supporting multilevel modelling to account for within-subject correlation.
```

```
#How the outcomes are nested within IDs (patients).
```

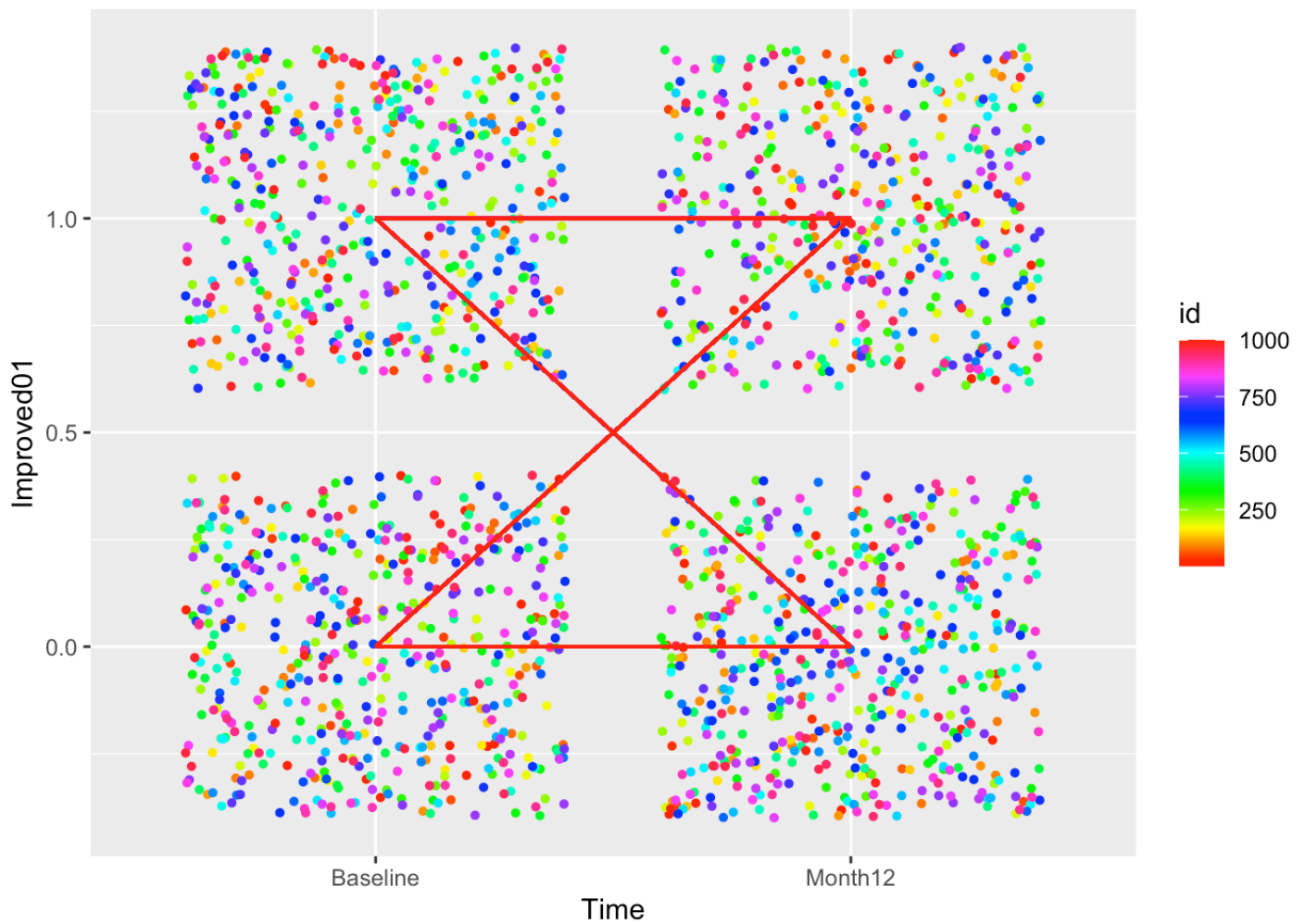
```
data$Improved01=ifelse(data$Improved=="Yes",1,0)
```

```
# Exploratory visualisation of repeated measurements
```

```
ggplot(data = data,  
       aes(x = Time,y = Improved01, col= id, group = id)) +  
  geom_jitter(size=1) +  
  geom_smooth(method = lm, se = FALSE, size = .6, alpha = .8) +  
  scale_color_gradientn(colours = rainbow(100))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

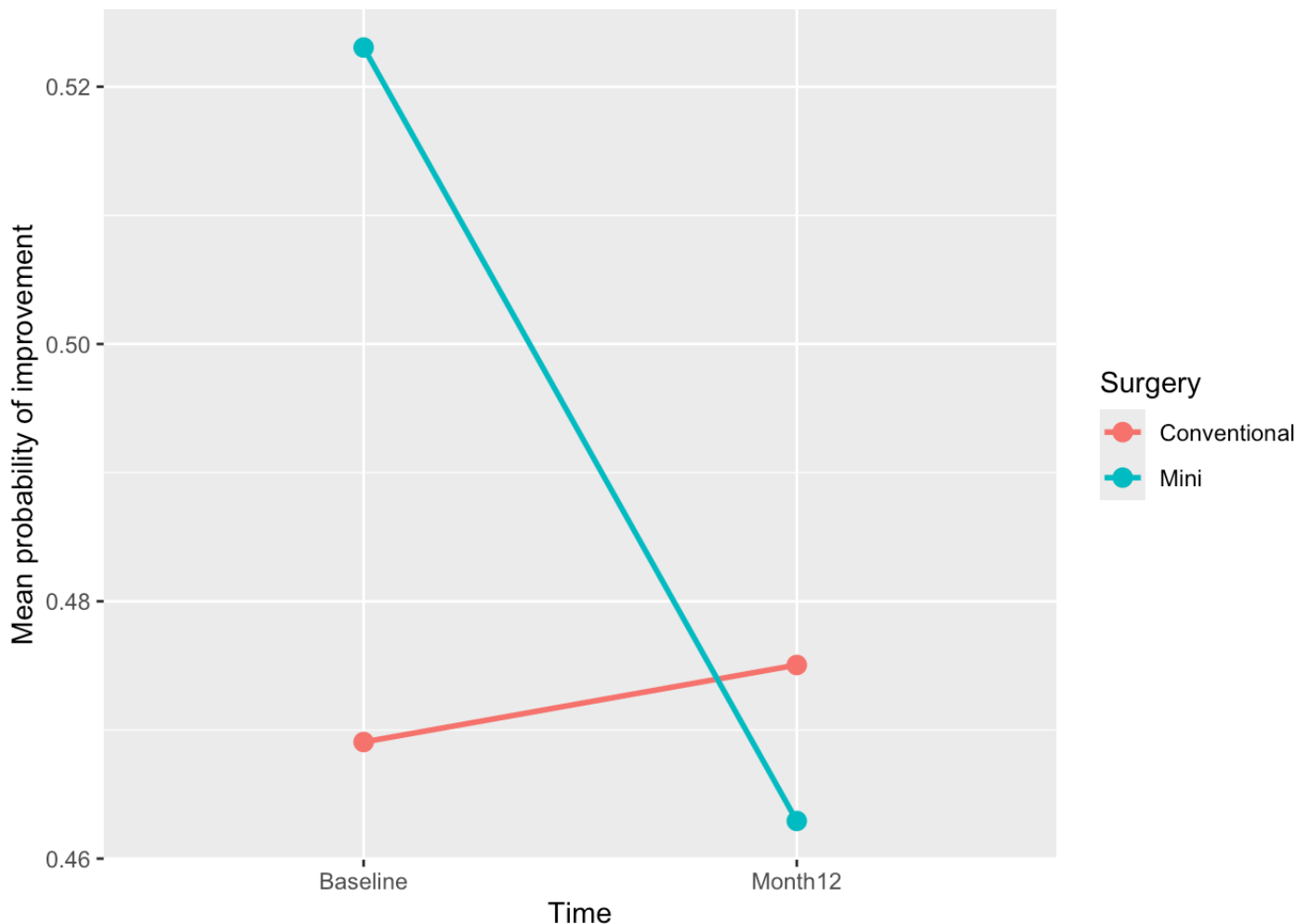
```
## `geom_smooth()` using formula = 'y ~ x'
```



Group-level mean trends

```
# Mean improvement rates by surgery and time
# Binary outcome => mean represents probability of improvement

ggplot(data, aes(Time, Improved01, group = Surgery, colour = Surgery)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  ylab("Mean probability of improvement") +
  xlab("Time")
```



Interpretation of within-patient longitudinal outcome plot

The plot shows individual patient outcomes at baseline and month 12, with each coloured trajectory representing repeated measurements from the same patient. The strong vertical clustering at 0 and 1 reflects the binary nature of the outcome, while the dense overlap of coloured points across time indicates substantial within-patient similarity alongside considerable between-patient heterogeneity.

The crossing fitted lines illustrate that changes over time vary markedly across individuals rather than following a consistent population-level trend. This visual evidence indicates that assuming independence would ignore important within-subject correlation, inflating variance and risking biased inference. Consequently, the data supports the use of a mixed-effects logistic regression model to account for patient-level clustering while avoiding overfitting through unnecessary fixed-effect complexity.

A key risk with longitudinal data is data leakage (same patient in train and test). To avoid this, we split by patient ID.

Model validation and predictive performance - Patient-level train-test split (to avoid information leakage) and **Confusion-matrix-based classification performance (test set)**

This code evaluates out-of-sample predictive performance using a patient-level 70/30 train-test split, ensuring all observations from the same patient are kept in one split to avoid information leakage. A logistic mixed-effects model (GLMM) with a random intercept for patient ID is fitted on the training data to account for within-patient correlation, and predictions for unseen patients are generated using fixed effects

only. Predictive performance is assessed using AUC (discrimination), the Brier score (calibration), and confusion-matrix metrics (accuracy, sensitivity, specificity, F1) at a 0.5 threshold. An equivalent logistic regression (GLM) is fitted for comparison, allowing assessment of whether accounting for patient-level heterogeneity improves generalisation and the bias–variance trade-off.

#The GLM corresponds to a standard logistic regression assuming independent observations, whereas the GLMM is a logistic mixed-effects model that includes a patient-level random intercept to account for repeated measurements over time.

```
set.seed(1)

ids=unique(data$id)
train_ids=sample(ids,size=floor(0.7*length(ids)))

train.data=data[data$id %in% train_ids,]
test.data =data[!data$id %in% train_ids,]

# GLMM (interaction)
m2_train=glmer(Improved01~Surgery*Time+(1|id),data=train.data,family=binomial)
p_test_glmm=predict(m2_train,newdata=test.data,type="response",allow.new.levels=TRUE)

#GLMM metrics
roc(test.data$Improved01,p_test_glmm)$auc
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Area under the curve: 0.5119
```

```
mean((p_test_glmm-test.data$Improved01)^2)
```

```
## [1] 0.2514385
```

```
pred_class_glmm=ifelse(p_test_glmm>=0.5,1,0)
```

```
# Confusion matrix for GLMM (threshold = 0.5)
test_pred_glmm = ifelse(p_test_glmm >= 0.5, 1, 0)

cm_glmm = confusionMatrix(
  factor(test_pred_glmm, levels = c(0,1)),
  factor(test.data$Improved01, levels = c(0,1)),
  positive = "1"
)

cm_glmm
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction    0    1
##              0 228 228
##              1  64  80
##
##              Accuracy : 0.5133
##              95% CI : (0.4725, 0.554)
##      No Information Rate : 0.5133
##      P-Value [Acc > NIR] : 0.5164
##
##              Kappa : 0.04
##
##  Mcnemar's Test P-Value : <2e-16
##
##      Sensitivity : 0.2597
##      Specificity : 0.7808
##      Pos Pred Value : 0.5556
##      Neg Pred Value : 0.5000
##      Prevalence : 0.5133
##      Detection Rate : 0.1333
##      Detection Prevalence : 0.2400
##      Balanced Accuracy : 0.5203
##
##      'Positive' Class : 1
##
```

```
cm_glmm$overall["Accuracy"]
```

```
## Accuracy
## 0.5133333
```

```
cm_glmm$byClass[c("Sensitivity", "Specificity", "Pos Pred Value", "Neg Pred Value", "F
1")]
```

##	Sensitivity	Specificity	Pos Pred Value	Neg Pred Value	F1
##	0.2597403	0.7808219	0.5555556	0.5000000	0.3539823

```
# GLM (interaction)
m_glm_train=glm(Improved01~Surgery*Time,family="binomial",data=train.data)
p_test_glm=predict(m_glm_train,newdata=test.data,type="response")

#GLM metrics
roc(test.data$Improved01,p_test_glm)$auc
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Area under the curve: 0.5119
```

```
mean((p_test_glm-test.data$Improved01)^2)
```

```
## [1] 0.2513444
```

```
pred_class_glm=ifelse(p_test_glm>=0.5,1,0)
```

```
# Confusion matrix for GLM (threshold = 0.5)
test_pred_glm = ifelse(p_test_glm >= 0.5, 1, 0)

cm_glm = confusionMatrix(
  factor(test_pred_glm, levels = c(0,1)),
  factor(test.data$Improved01, levels = c(0,1)),
  positive = "1"
)

cm_glm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 228 228
##           1  64  80
##
##           Accuracy : 0.5133
##           95% CI : (0.4725, 0.554)
##           No Information Rate : 0.5133
##           P-Value [Acc > NIR] : 0.5164
##
##           Kappa : 0.04
##
## Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.2597
##           Specificity : 0.7808
##           Pos Pred Value : 0.5556
##           Neg Pred Value : 0.5000
##           Prevalence : 0.5133
##           Detection Rate : 0.1333
##           Detection Prevalence : 0.2400
##           Balanced Accuracy : 0.5203
##
##           'Positive' Class : 1
##
```

```
cm_glm$overall["Accuracy"]
```

```
## Accuracy
## 0.5133333
```

```
cm_glm$byClass[c("Sensitivity","Specificity","Pos Pred Value","Neg Pred Value","F1")]
```

##	Sensitivity	Specificity	Pos Pred Value	Neg Pred Value	F1
##	0.2597403	0.7808219	0.5555556	0.5000000	0.3539823

Interpretation of results

The mitral valve improvement outcome was analysed using a logistic mixed-effects model with a random intercept for patient ID, which is appropriate for repeated binary measurements at baseline and 12 months and the resulting within-patient correlation. Fixed effects for surgery type, time, and their interaction were small and not statistically significant, suggesting no clear evidence that average improvement differs between surgical approaches or changes materially over time. Predictive performance assessed via a patient-level 70/30 train-test split showed very weak discrimination (AUC $\approx$ 0.51) and limited overall

probabilistic accuracy (Brier ≈ 0.25). At a 0.5 threshold, confusion-matrix results indicated accuracy close to the no-information rate, low sensitivity (~ 0.26 – 0.27) but moderate specificity (~ 0.78 – 0.79), meaning the model identifies “no improvement” better than “improvement.”

A standard logistic regression produced almost identical AUC/Brier and confusion-matrix metrics, implying that while the mixed-effects model is the correct choice for valid longitudinal inference, accounting for patient-level heterogeneity does not materially improve out-of-sample prediction when using Surgery and Time alone. Overall, these findings are consistent with limited predictive signal (high-bias/underfitting) rather than overfitting or instability across splits.

Varying training size (training vs test error)

This code estimates a learning-curve style diagnostic by repeatedly fitting the same logistic regression model on progressively larger patient-level training sets and computing both training and test error rates at each size. Plotting training vs test error across training sizes diagnoses the bias–variance trade-off: consistently high errors indicate high bias/underfitting, a large gap between training and test error indicates overfitting/high variance, and unstable test error as training size changes signals sensitivity to sampling (variance).

```

# Bias/Variance check

acc = NULL # Store accuracies
ids = unique(data$id) # Store unique patient IDs
n_ids = length(ids) # Total number of unique patients

for (i in seq(50, n_ids - 50, by = 50)) { # Vary training size from 50 to (total - 50) in increments of 50

  train_ids = sample(ids, i) # Randomly sample i patient IDs for training
  train.data = data[data$id %in% train_ids, ] # Training data
  test.data = data[!data$id %in% train_ids, ] # Test data

  if (nrow(test.data) == 0) next # Skip if no test data

  m = glm(Improved01 ~ Surgery + Time, # Fit logistic regression model
          family = "binomial",
          data = train.data)

  pred_test = predict(m, newdata = test.data, type = "response") # Predict on test data
  pred_class_test = ifelse(pred_test >= 0.5, 1, 0) # Classify predictions
  acc.test = mean(pred_class_test == test.data$Improved01) # Test accuracy

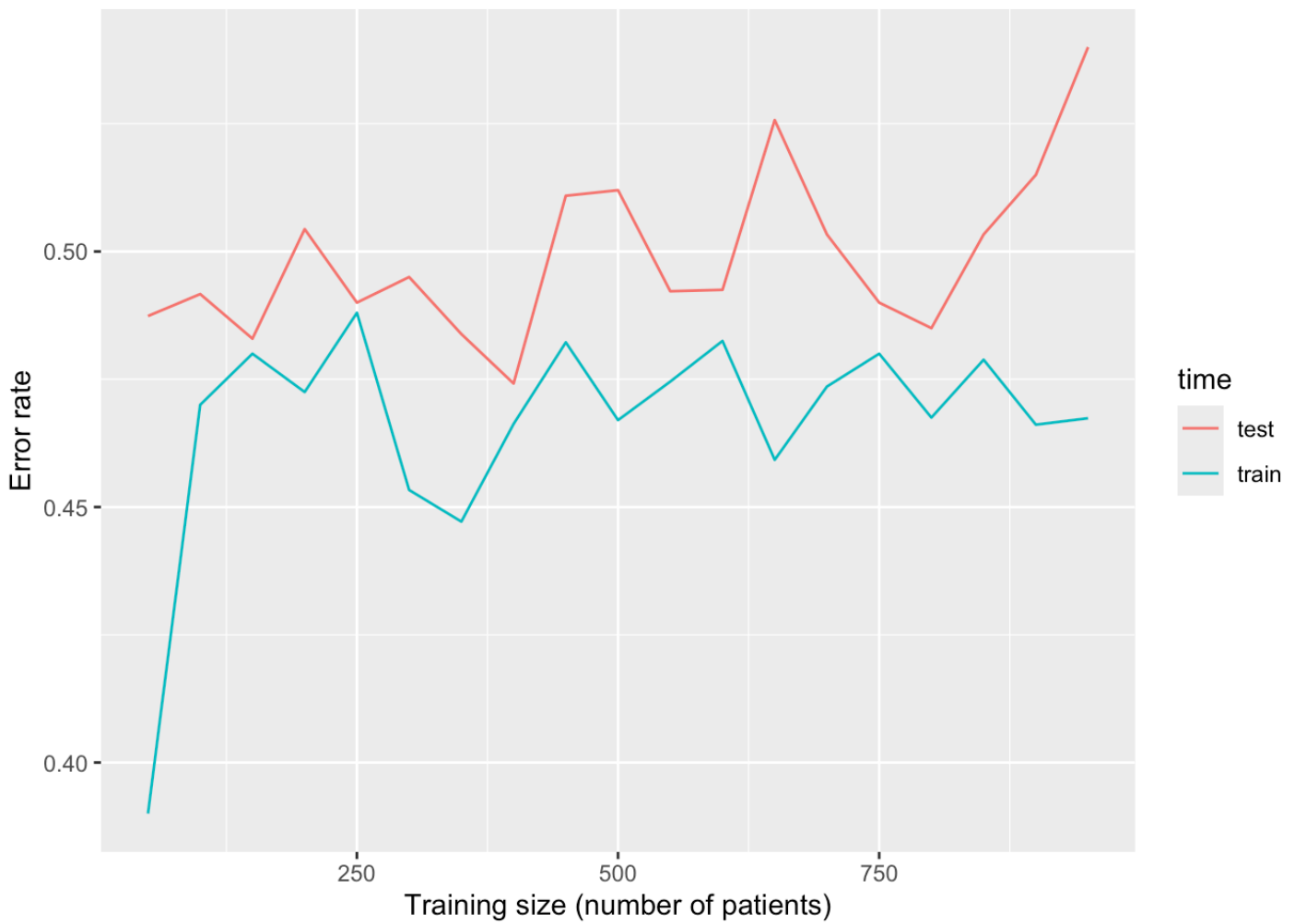
  pred_train = predict(m, newdata = train.data, type = "response") # Predict on training data
  pred_class_train = ifelse(pred_train >= 0.5, 1, 0) # Classify predictions
  acc.train = mean(pred_class_train == train.data$Improved01) # Training accuracy

  acc = rbind(acc, # Combine accuracies into a data frame
              data.frame(acc.test = acc.test,
                          acc.train = acc.train,
                          train.size = i))
}

acc = reshape(acc, varying = 1:2, direction = "long") # Reshape for plotting

ggplot(acc) + # Plot error rates vs training size
  geom_line(aes(train.size, 1 - acc, col = time)) +
  xlab("Training size (number of patients)") +
  ylab("Error rate")

```



Training error vs test error across training sizes indicates underfitting/overfitting patterns and variance across samples.

Interpretation of training vs test error across training sizes

The learning curve shows that both training and test error remain relatively high and close to each other across all training sizes, with no substantial reduction as more patients are added. This pattern indicates high bias (underfitting), as the model is unable to capture sufficient structure in the data even with increased training data. The small and stable gap between training and test error suggests that overfitting is limited and variance is not excessive; however, the noticeable fluctuations in test error across training sizes indicate some sampling variability, reflecting residual variance driven by unmodelled patient heterogeneity. Overall, the plot suggests that improving performance would require additional or more informative predictors rather than increased model complexity.

Fixed training size, different splits (test error instability)

This code repeatedly refits the same logistic regression model across 100 different patient-level 70/30 splits and plots the resulting test error each time. The purpose is to quantify variance in predictive performance: large fluctuations indicate instability and sensitivity to which patients are sampled (high variance), while consistently high test error across iterations indicates persistent bias (underfitting) rather than overfitting to a particular split.

```

acc=NULL # Store accuracies
ids=unique(data$id) # set of unique patient IDs

for (i in 1:100){

  train_ids = sample(ids, size = floor(0.7 * length(ids))) # 70% train
  train.data=data[data$id %in% train_ids,] # Training data
  test.data =data[!data$id %in% train_ids,] # Test data

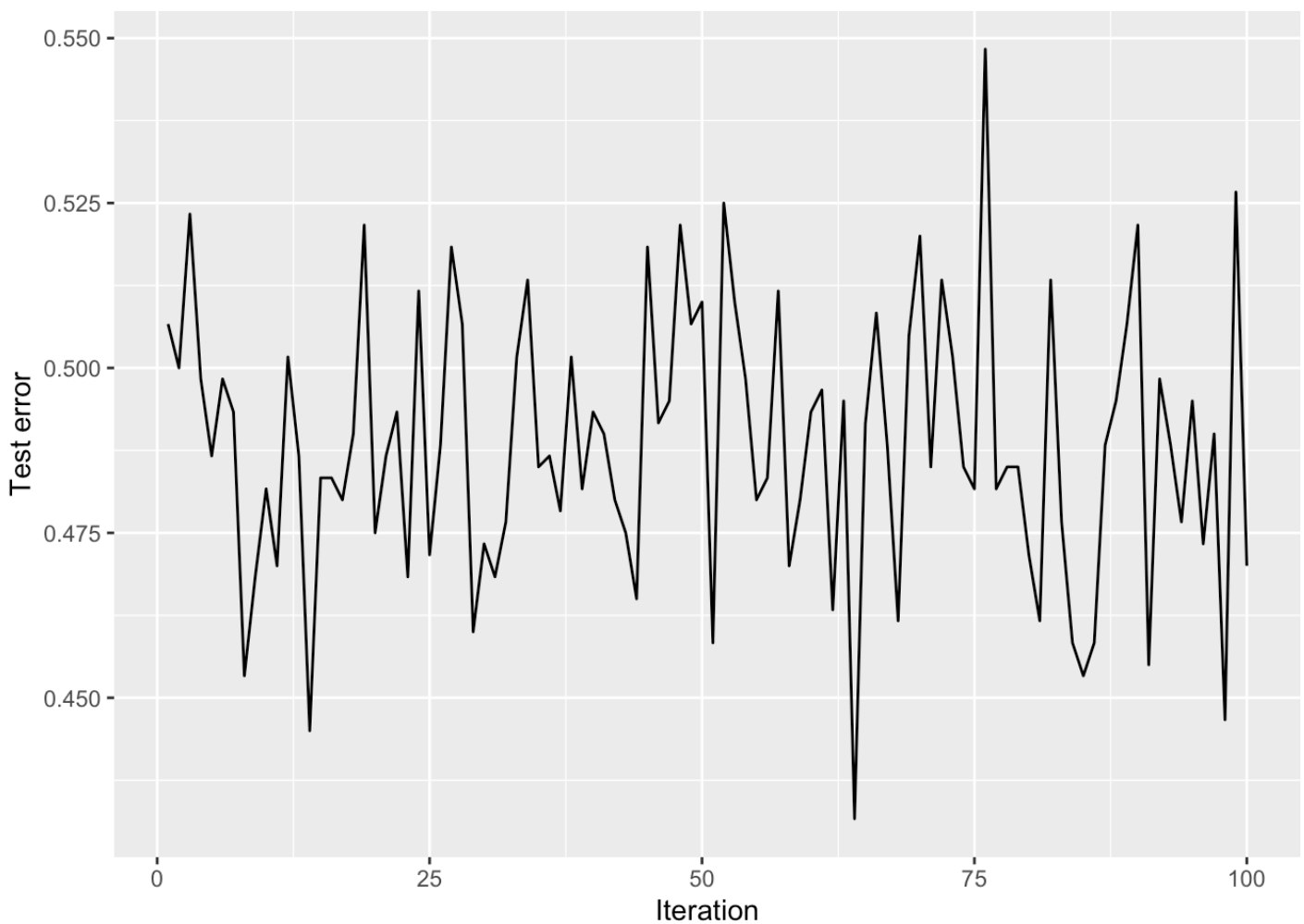
  m=glm(Improved01~Surgery+Time,family="binomial",data=train.data) # Fit model

  pred=predict(m,newdata=test.data,type="response") # Predict on test data
  pred_class=ifelse(pred>=0.5,1,0) # Classify predictions

  acc=c(acc,mean(pred_class==test.data$Improved01)) # Store test accuracy
}

ggplot() + geom_line(aes(1:length(acc),1-acc)) +
  xlab("Iteration") + ylab("Test error")

```



```
# Test error varies across repeated random splits, showing sampling variability and modest predictive performance.
```

Interpretation of test-error instability across repeated splits

The plot shows noticeable variation in test error across repeated random patient-level splits, indicating sensitivity of model performance to which patients are included in the training set. This fluctuation reflects sampling variance driven by heterogeneity between patients, rather than overfitting to a particular split, as test error remains consistently high rather than collapsing toward zero. The absence of a systematic downward trend in error further indicates persistent bias (underfitting), suggesting that surgery and time alone provide limited predictive signal. Overall, the instability confirms moderate variance in generalisation performance while reinforcing that the dominant limitation of the model is structural bias rather than excessive complexity.

ROC curves and AUC comparison (GLMER vs GLM)

This code evaluates and compares the predictive performance of the logistic mixed-effects model (GLMM) and the standard logistic regression model (GLM) on unseen patients. ROC curves and AUC are used to assess how well each model discriminates between improvement and no improvement, independent of any classification threshold. Threshold tuning then explores whether changing the probability cut-off improves classification metrics such as accuracy and F1-score. Overall, the procedure helps determine whether performance differences are due to the modelling approach or simply the choice of decision threshold, and whether poor performance reflects underfitting rather than overfitting.

```
# ROC curves assess discrimination (ranking ability of predicted probabilities)
# AUC close to 0.5 indicates weak discrimination / high bias

# ---- ROC + AUC ----
roc_glmm = roc(test.data$Improved01, p_test_glmm) # ROC for mixed-effects model
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
roc_glm = roc(test.data$Improved01, p_test_glm) # ROC for standard logistic regression
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

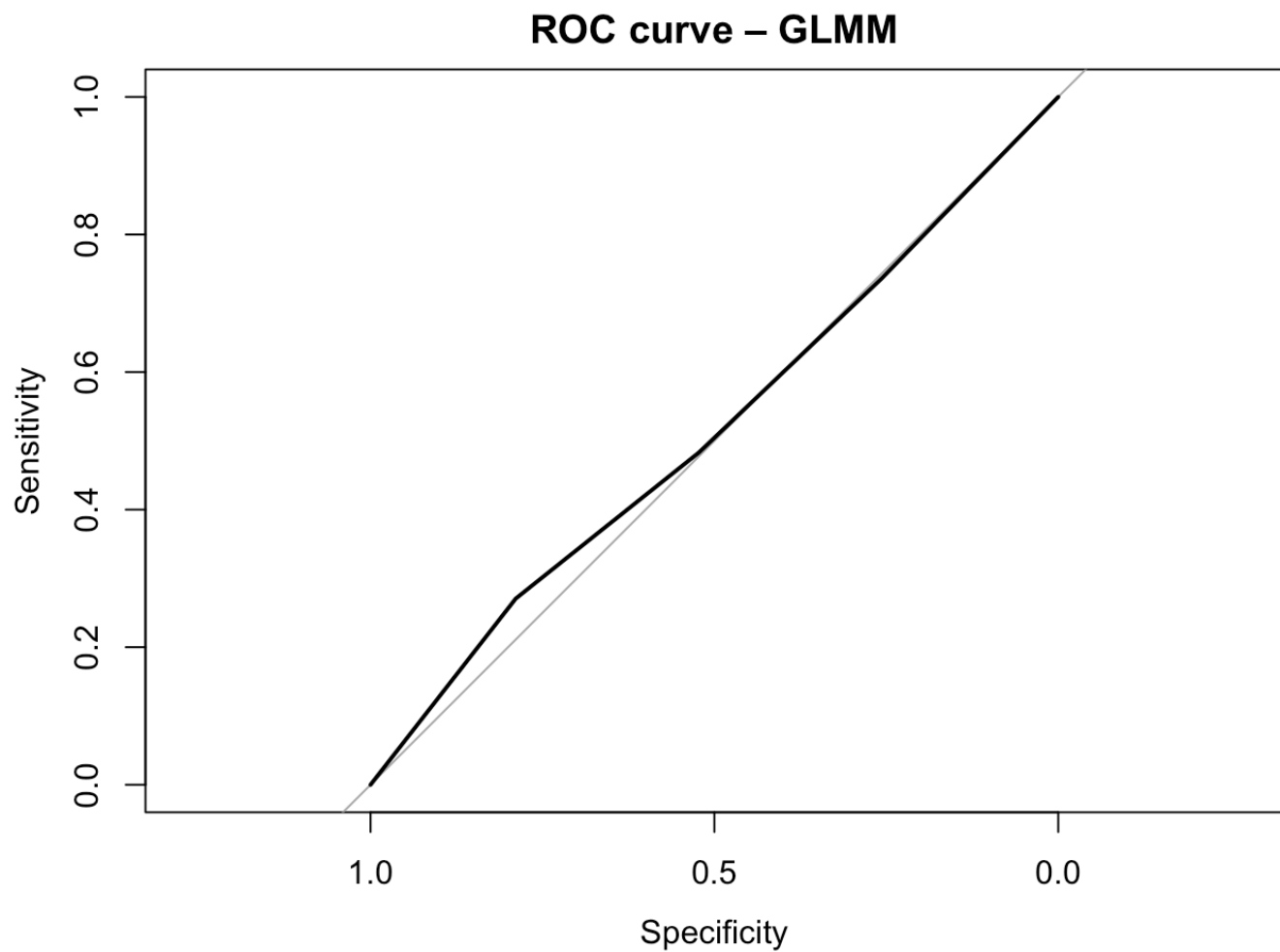
```
auc(roc_glmm) # AUC for GLMM
```

```
## Area under the curve: 0.5138
```

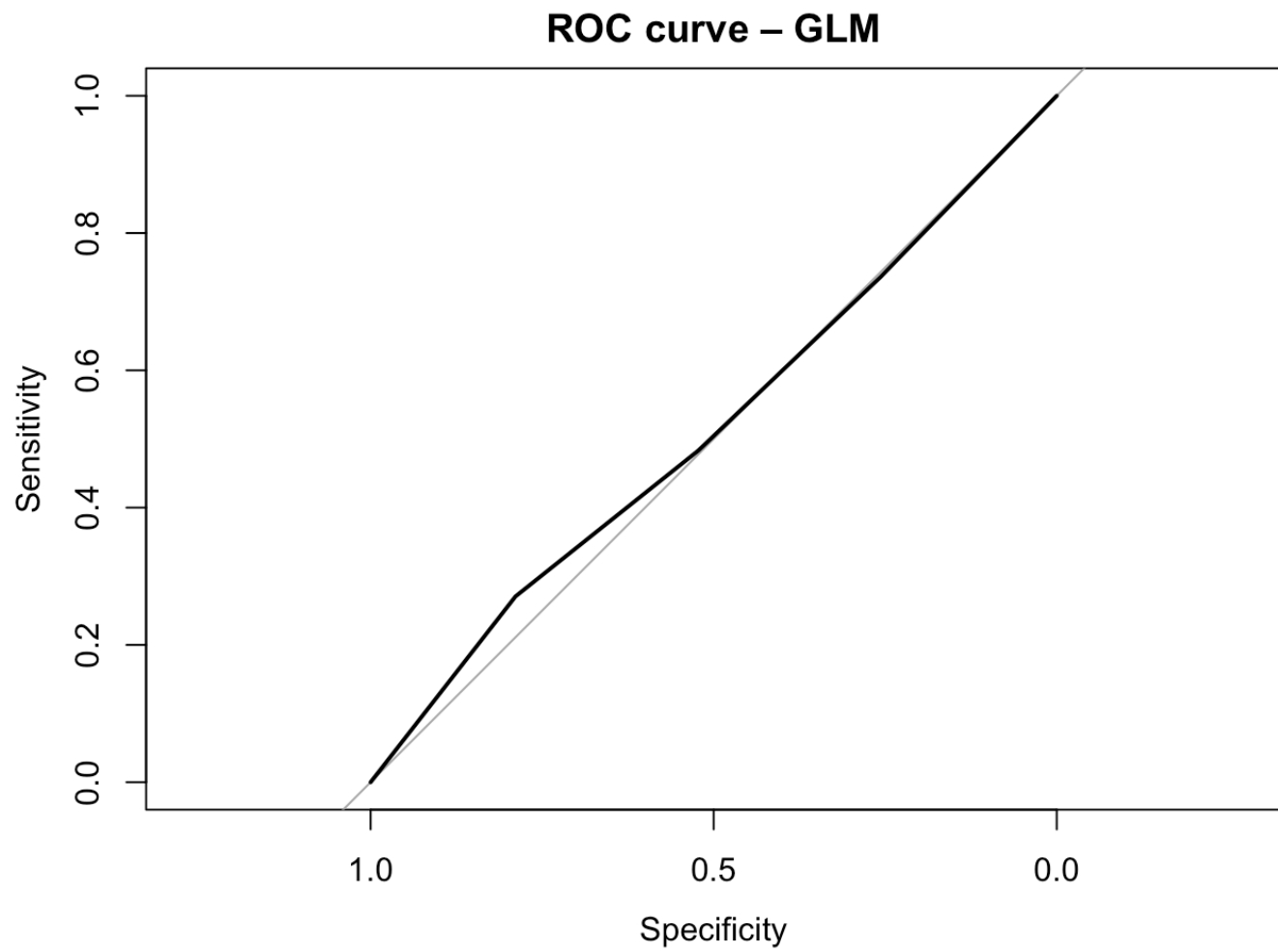
```
auc(roc_glm)    # AUC for GLM
```

```
## Area under the curve: 0.5138
```

```
plot(roc_glmm, main = "ROC curve – GLMM")  # ROC plot for GLMM
```



```
plot(roc_glm, main = "ROC curve – GLM")    # ROC plot for GLM
```



```

# ---- Threshold tuning (grid search) ----
# Tests how classification performance changes with different probability cut-offs

thresholds = seq(0.05, 0.95, by = 0.05) # Range of classification thresholds

tune_glmm = NULL
for (t in thresholds){
  pred_t = ifelse(p_test_glmm >= t, 1, 0) # Convert probabilities to class labels
  cm_t = confusionMatrix(
    factor(pred_t, levels = c(0,1)),
    factor(test.data$Improved01, levels = c(0,1)),
    positive = "1"
  )
  tune_glmm = rbind(tune_glmm,
                    data.frame(threshold = t,
                               Accuracy = as.numeric(cm_t$overall["Accuracy"]),
                               F1 = as.numeric(cm_t$byClass["F1"]),
                               model = "GLMM"))
}

tune_glm = NULL
for (t in thresholds){
  pred_t = ifelse(p_test_glm >= t, 1, 0) # Thresholded predictions for GLM
  cm_t = confusionMatrix(
    factor(pred_t, levels = c(0,1)),
    factor(test.data$Improved01, levels = c(0,1)),
    positive = "1"
  )
  tune_glm = rbind(tune_glm,
                   data.frame(threshold = t,
                              Accuracy = as.numeric(cm_t$overall["Accuracy"]),
                              F1 = as.numeric(cm_t$byClass["F1"]),
                              model = "GLM"))
}

```

```

tune_all = rbind(tune_glmm, tune_glm) # Combine results for comparison

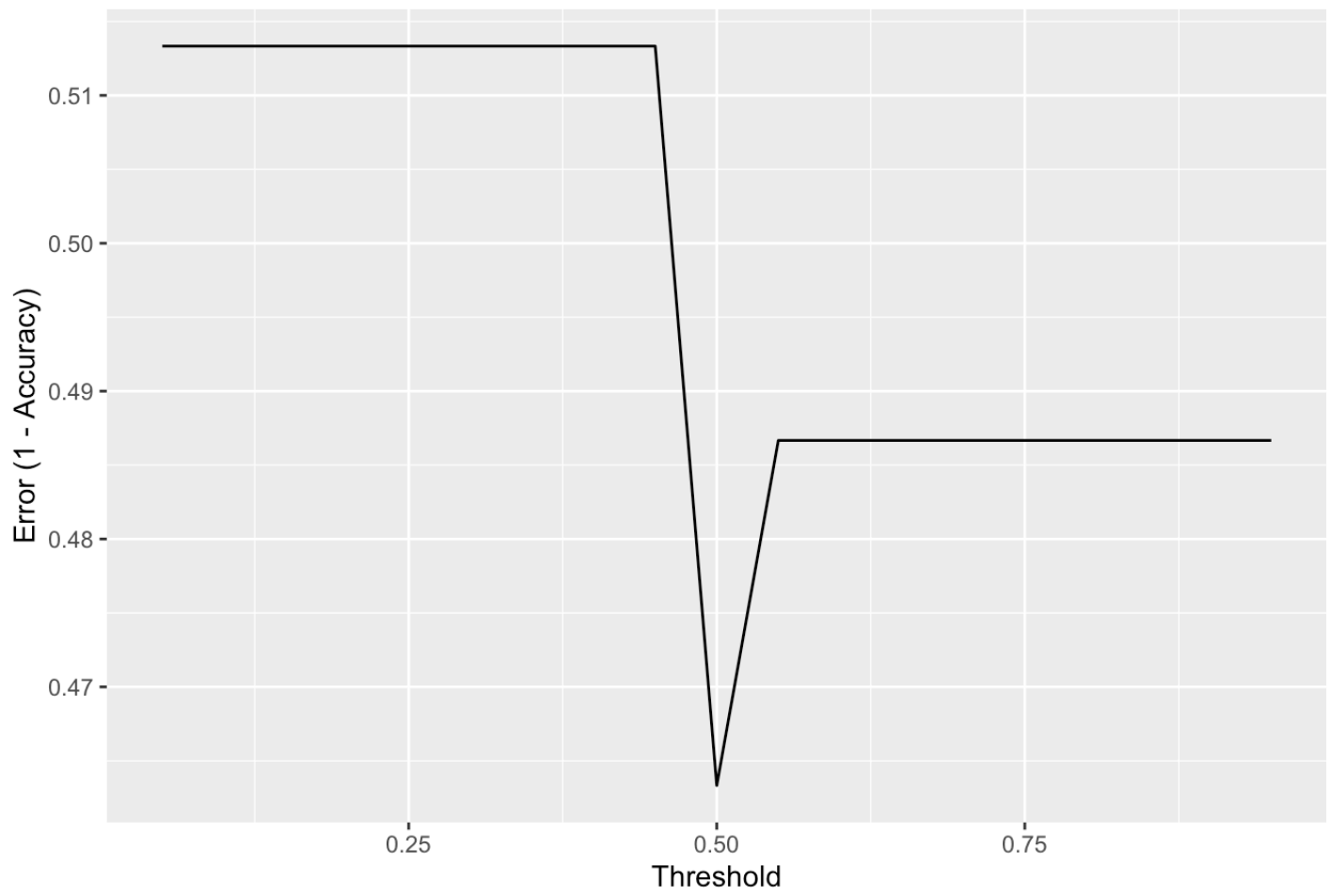
# Display thresholds with highest F1 scores
tune_all[order(-tune_all$F1), ][1:20, ]

```

##	threshold	Accuracy	F1	model
## 1	0.05	0.4866667	0.6547085	GLMM
## 2	0.10	0.4866667	0.6547085	GLMM
## 3	0.15	0.4866667	0.6547085	GLMM
## 4	0.20	0.4866667	0.6547085	GLMM
## 5	0.25	0.4866667	0.6547085	GLMM
## 6	0.30	0.4866667	0.6547085	GLMM
## 7	0.35	0.4866667	0.6547085	GLMM
## 8	0.40	0.4866667	0.6547085	GLMM
## 9	0.45	0.4866667	0.6547085	GLMM
## 20	0.05	0.4866667	0.6547085	GLM
## 21	0.10	0.4866667	0.6547085	GLM
## 22	0.15	0.4866667	0.6547085	GLM
## 23	0.20	0.4866667	0.6547085	GLM
## 24	0.25	0.4866667	0.6547085	GLM
## 25	0.30	0.4866667	0.6547085	GLM
## 26	0.35	0.4866667	0.6547085	GLM
## 27	0.40	0.4866667	0.6547085	GLM
## 28	0.45	0.4866667	0.6547085	GLM
## 10	0.50	0.5366667	0.3623853	GLMM
## 29	0.50	0.5366667	0.3623853	GLM

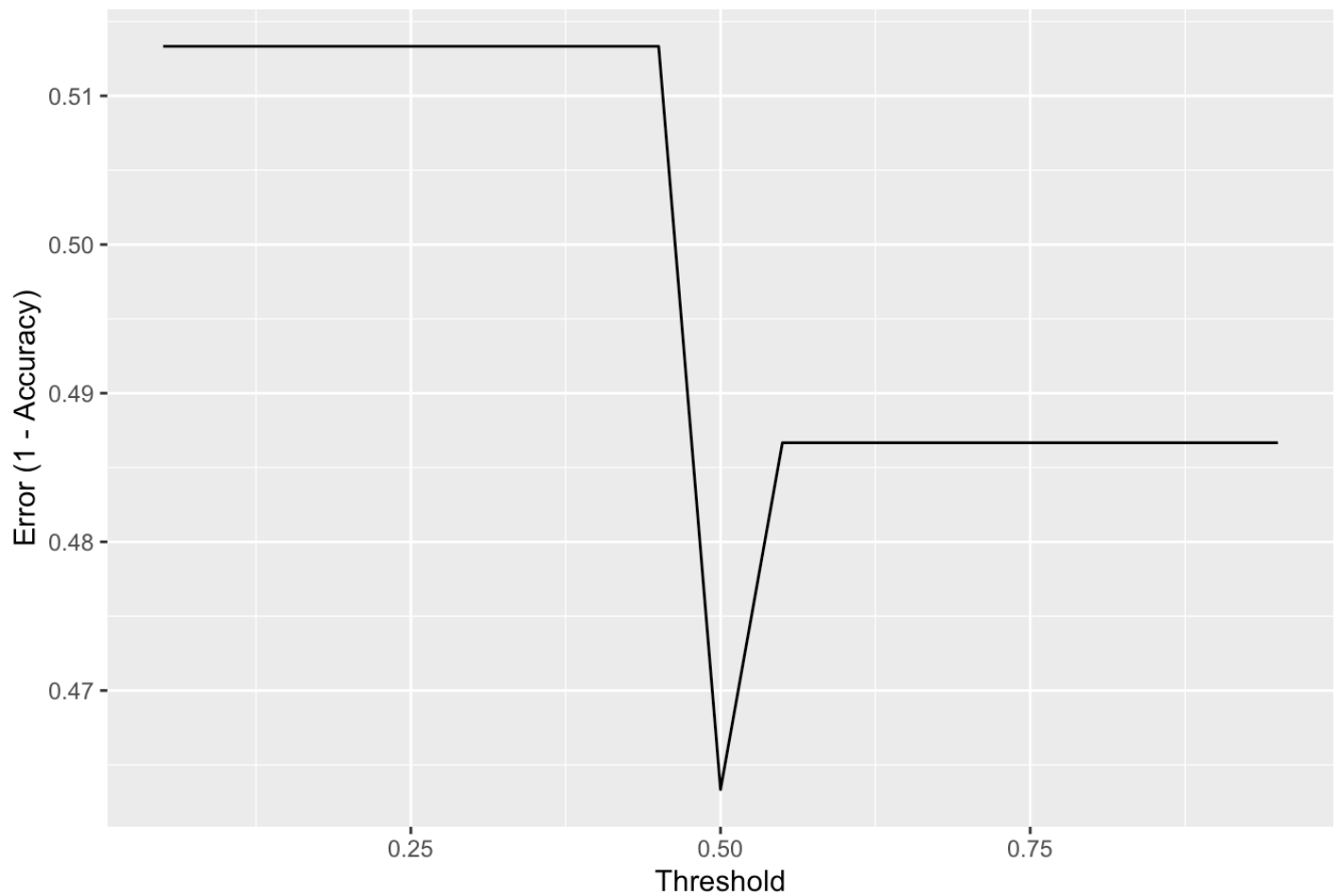
```
# Plot classification error vs threshold
ggplot(tune_glmm) +
  geom_line(aes(threshold, 1 - Accuracy)) +
  xlab("Threshold") + ylab("Error (1 - Accuracy)") +
  ggtitle("Threshold tuning - GLMM")
```

Threshold tuning – GLMM



```
ggplot(tune_glm) +  
  geom_line(aes(threshold, 1 - Accuracy)) +  
  xlab("Threshold") + ylab("Error (1 - Accuracy)") +  
  ggtitle("Threshold tuning – GLM")
```

Threshold tuning – GLM



```
# Select thresholds that maximise F1 score
best_t_glmm = tune_glmm$threshold[which.max(tune_glmm$F1)]
best_t_glm  = tune_glm$threshold[which.max(tune_glm$F1)]

best_t_glmm
```

```
## [1] 0.05
```

```
best_t_glm
```

```
## [1] 0.05
```

```
# Final predictions using best thresholds
glmm_best_pred = ifelse(p_test_glmm >= best_t_glmm, 1, 0)
glm_best_pred  = ifelse(p_test_glm  >= best_t_glm, 1, 0)

# Confusion matrices at best thresholds
confusionMatrix(
  factor(glmm_best_pred, levels = c(0,1)),
  factor(test.data$Improved01, levels = c(0,1)),
  positive = "1"
)
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction    0    1
##              0    0    0
##              1 308 292
##
##              Accuracy : 0.4867
##              95% CI : (0.446, 0.5275)
##      No Information Rate : 0.5133
##      P-Value [Acc > NIR] : 0.9111
##
##              Kappa : 0
##
##  McNemar's Test P-Value : <2e-16
##
##      Sensitivity : 1.0000
##      Specificity : 0.0000
##      Pos Pred Value : 0.4867
##      Neg Pred Value :    NaN
##      Prevalence : 0.4867
##      Detection Rate : 0.4867
##      Detection Prevalence : 1.0000
##      Balanced Accuracy : 0.5000
##
##      'Positive' Class : 1
##
```

```
confusionMatrix(
  factor(glm_best_pred, levels = c(0,1)),
  factor(test.data$Improved01, levels = c(0,1)),
  positive = "1"
)
```

```

## Confusion Matrix and Statistics
##
##               Reference
## Prediction    0    1
##              0    0    0
##              1  308  292
##
##               Accuracy : 0.4867
##               95% CI : (0.446, 0.5275)
##       No Information Rate : 0.5133
##       P-Value [Acc > NIR] : 0.9111
##
##               Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##       Sensitivity : 1.0000
##       Specificity : 0.0000
##       Pos Pred Value : 0.4867
##       Neg Pred Value :    NaN
##       Prevalence : 0.4867
##       Detection Rate : 0.4867
##       Detection Prevalence : 1.0000
##       Balanced Accuracy : 0.5000
##
##       'Positive' Class : 1
##

```

Interpretation

The results show that both the logistic mixed-effects model (GLMM) and the standard logistic regression (GLM) have very weak predictive performance for mitral valve improvement. The ROC curves for both models lie close to the diagonal, with identical AUC values of approximately 0.51, indicating discrimination no better than random guessing. Confusion matrices at the default 0.5 threshold show accuracy close to the no-information rate, with low sensitivity and higher specificity, meaning the models are better at identifying patients with no improvement than those who do improve. Threshold tuning changes accuracy and F1 scores, but the best F1 values occur at very low thresholds where almost all patients are predicted to improve, leading to perfect sensitivity but zero specificity, which does not represent meaningful improvement. The near-identical ROC curves, threshold-error plots, and confusion matrices for GLM and GLMM indicate that accounting for patient-level random effects is necessary for valid longitudinal inference but does not improve out-of-sample prediction when using Surgery and Time alone. Overall, the results consistently point to underfitting caused by limited predictive signal rather than overfitting or model instability.

Conclusion and Suggestions

This study analysed mitral valve improvement using both standard logistic regression (GLM) and logistic mixed-effects regression (GLMM) to account for the longitudinal structure of the data. The GLMM was methodologically more appropriate because it explicitly modelled within-patient correlation through a random intercept, whereas the GLM assumed independence between repeated observations. However, both models produced almost identical predictive performance, with AUC values close to 0.5 and classification accuracy near the no-information rate. This indicates that although the mixed-effects model is required for valid inference, accounting for patient-level heterogeneity did not improve out-of-sample prediction when only surgery type and time were included as predictors.

Challenges and limitations

A key challenge identified was the limited predictive signal in the available features. Surgery type and time alone explained very little variation in the outcome, leading to persistent underfitting across all models. This issue was partially addressed by using appropriate longitudinal modelling, patient-level train–test splitting to avoid data leakage, and multiple validation techniques (ROC/AUC, Brier score, confusion matrices, learning curves, and repeated splits). However, the lack of additional informative covariates meant that the core limitation of structural bias could not be fully addressed. Threshold tuning improved some classification metrics superficially, but resulted in unrealistic decision rules with sensitivity and zero specificity, and therefore did not represent genuine performance gains.

Suggestions for future work

Future work could improve predictive performance by incorporating additional clinically relevant predictors such as age, disease severity, comorbidities, or continuous clinical measurements. Longer follow-up periods with more than two time points may also better capture individual recovery trajectories. Alternative modelling approaches, such as random-slope mixed models or joint models combining longitudinal and survival outcomes, could be explored. Finally, external validation on an independent dataset would help assess generalisability and robustness of the findings.

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